

Normal Distribution, Pt 1

CSU, Chico Math 350

2026-04-24

outline

Basics

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Situation

The most widely used continuous distribution is the normal distribution, a distribution with the familiar “bell” shape. Many characteristics in nature exhibit this shape:

- ▶ Heights of humans, trees, capybaras, etc.
- ▶ Failure of mechanical parts due to wear and tear
- ▶ Random noise in electrical circuits

Situation

The normal distribution is an extremely important distribution and we will discuss some of its properties in this section. There are three main reasons why the normal distribution is so important:

- ▶ Many random variables happen to have a normal distribution. Other times, a transformation of a random variable has a normal distribution.
- ▶ The **central limit theorem** states that under certain conditions the distribution of the sample mean, $\bar{x}$, will be normally distributed.
- ▶ It is a mathematical convenience to be able to assume that the distribution from which a sample is drawn is a normal distribution.

Random variable: Let X be a random variable from a Normal distribution defined by the parameters μ for the mean, and σ^2 for the variance.

Normal Distribution Properties

Distributional Notation

$$X \sim N(\mu, \sigma^2)$$

PDF

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mean & Variance

$$E(X) = \mu$$

$$\text{Var}(X) = \sigma^2$$

R Commands: (R uses *standard deviation*, NOT the variance)

`dnorm(x, mu, sd)` to compute $P(X = x)$ (height of curve)

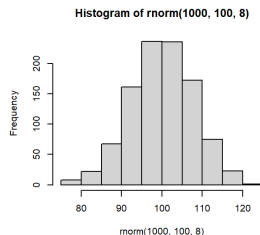
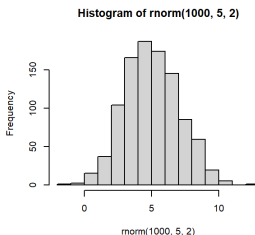
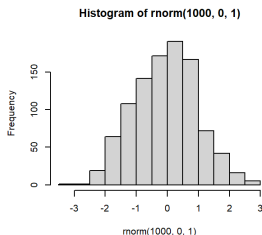
`pnorm(x, mu, sd)` to compute $P(X \leq x)$ (the CDF)

`rnorm(N, mu, sd)` to randomly draw N samples from a

$X \sim N(\mu, \sigma^2)$ distribution.

Visualizing the Shape

```
par(mfrow=c(1,3))  
hist(rnorm(1000, 0, 1))  
hist(rnorm(1000, 5, 2))  
hist(rnorm(1000, 100, 8))
```



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Defining Z

The **standard normal random variable** Z is a special case of the Normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$. The PDF then simplifies to:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

This is also commonly known as the “Z Distribution”, and is a transformation of the random variable X :

$$Z = \frac{X - \mu}{\sigma}$$

Normalization

This is also known as “normalizing” (or standardizing) a variable. If you have any random variable you can subtract the mean (center on zero), and divide by its standard deviation (scaling) to achieve a Standard Normal, or Z distribution.

This is a very useful way to compare values between two random variables with different means and/or variances. You transform them both to Z-distributions.

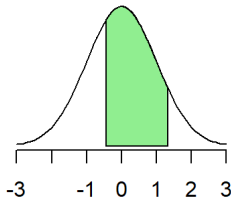
Once your two random variables have the same mean and variance, they're much easier to compare!

Example 1

Use R to evaluate the following integrals under the Standard Normal Z distribution. In each case, draw a diagram of $f_Z(z)$ and shade the area that corresponds to the integral.

$$A) \frac{1}{\sqrt{2\pi}} \int_{-.44}^{1.33} e^{-z^2/2}$$

```
ex1 <- function(z){1/sqrt(2*pi)*exp(-(z^2)/2)}  
integrate(ex1, -.44, 1.33)
```

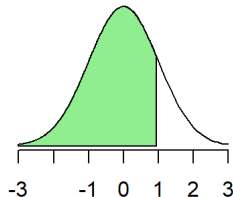


Example 1

Use R to evaluate the following integrals under the Standard Normal Z distribution. In each case, draw a diagram of $f_Z(z)$ and shade the area that corresponds to the integral.

$$\text{B) } \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0.94} e^{-z^2/2}$$

```
ex1 <- function(z){1/sqrt(2*pi)*exp(-(z^2)/2)}  
integrate(ex1, -Inf, 0.94) # R reads Inf as infinity
```



x

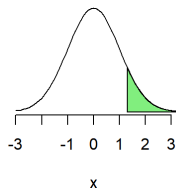
Example 2

Use `pnorm` to calculate the theoretical probability for each question, and confirm via simulation using `rnorm`.

A) $P(Z > 1.3)$

Theoretical

```
1-pnorm(1.3)  
## [1] 0.09680048
```



Simulation

```
z <- rnorm(10000, 0, 1)  
mean(z>1.3)  
## [1] 0.0974
```

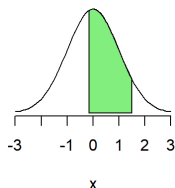
Example 2

Use `pnorm` to calculate the theoretical probability for each question, and confirm via simulation using `rnorm`.

$$\text{B) } P(-0.15 < Z < 1.5)$$

Theoretical

```
pnorm(1.5)-pnorm(-0.15)  
## [1] 0.4928105
```



Simulation

```
z <- rnorm(10000, 0, 1)  
mean(z > -0.15 & z < 1.5)  
## [1] 0.4878
```

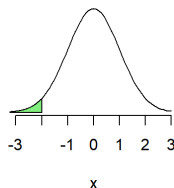
Example 2

Use `pnorm` to calculate the theoretical probability for each question, and confirm via simulation using `rnorm`.

C) $P(Z < -2)$

Theoretical

```
pnorm(-2)
## [1] 0.02275013
```



Simulation

```
z <- rnorm(10000, 0, 1)
mean(z < -2)
## [1] 0.0231
```

Exercise 1

Use R for all steps—do not do these by hand or using Z tables. Use the `integrate`, `pnorm`, and `rnorm` functions.

1. $\frac{1}{\sqrt{2\pi}} \int_{-1}^2 e^{-z^2/2}$

2. $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{2.1} e^{-z^2/2}$

3. $P(Z < 0.9)$

4. $P(1.1 < Z < 2.5)$

5. $P(Z > 0.9)$

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.